

ML Development Workflow Example

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ML Workshop Example

Problem at Hand: Counting and Trait Extraction of Blueberries

Steps:

- Understanding the Data and Problem
- Formalizing the Tasks
- Data Preparation and Annotation:
 - Selecting a Schema
 - Hierarchical Annotation
 - AI Assisted Annotation
- Developing and Evaluating the Model
- Deploying the Model



AI Collaboration Modes for Content Generation

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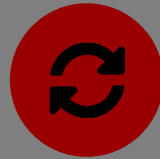
Three Modes of Working with an AI Code Assistant

Cyborgs, Centaurs, and Self-Automators — from a Harvard Business School study of AI-assisted knowledge work



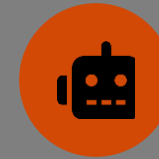
Centaur

A clean division of labor. You decide what needs to happen, then delegate one bounded piece to the AI — and take control back to verify it.



Cyborg

Continuous interleaving, not a clean split. You and the AI go back and forth repeatedly within the same task — no clear line between “your part” and “its part.”



Self-Automator

The whole task goes to the AI. One high-level prompt, minimal ongoing engagement, review only at the end.

Let's think of examples of use on non-coding tasks

Same ML Task, Three Different Workflows

What each mode looks like when building a model with an AI code assistant

	Centaur	Cyborg	Self-Automator
Your role	<i>Decide architecture & strategy yourself</i>	<i>Co-write live — start, edit, hand off, repeat</i>	<i>Write one high-level prompt</i>
AI’s role	<i>Handles one bounded, delegated task</i>	<i>Continuously drafts, edits, explains errors with you</i>	<i>Runs the pipeline end-to-end, largely unattended</i>
Example prompt	<i>“Write the training loop for this architecture”</i>	<i>Live back-and-forth fixing a failing test together</i>	<i>“Build a model that predicts X from this dataset”</i>

None of these is “the right way” — the point is recognizing which mode you’re in and choosing it on purpose.

Designing an ML Workflow

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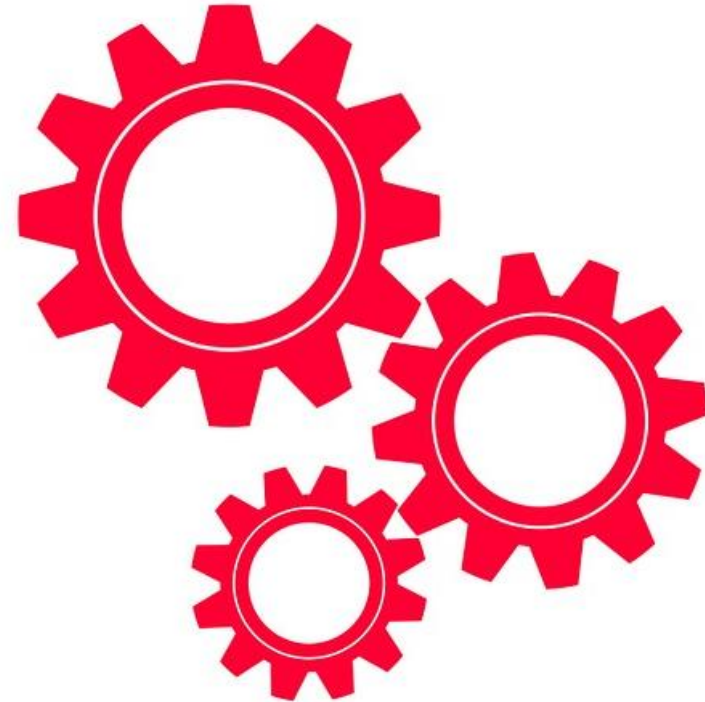
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The Problem: Synthetic Gait Classification

- We are aiming to develop a personalized model that determine when a person is walking fast or slow. We have conducted two recording sessions, in which the individual switches between fast and slow walking periodically. Let's build a model that predicts which gait it is observed in the data.

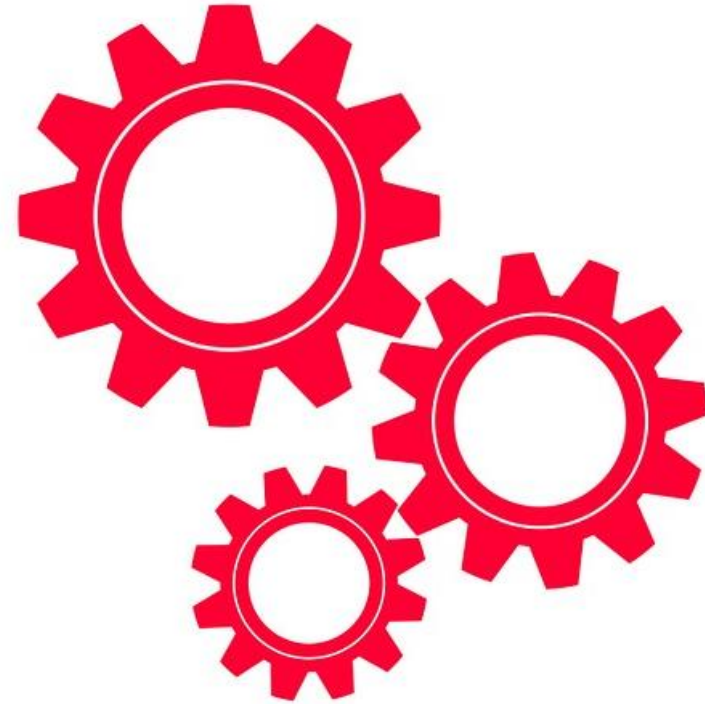
The Problem: Synthetic Gait Classification

- Let's do this as a Self-Automator



The Problem: Synthetic Gait Classification

- Let's do this as a Centaur



The Problem: Synthetic Gait Classification

- Let's refine as a Cyborg

